

Algebra Revision

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29 September 2026

1 Audit

1.1 Indices

1. Simplify:

a)

$$\begin{aligned}2^5 \cdot 2^{-3} \div 2^4 &= 2^{5-3-4} \\ &= 2^{-2}\end{aligned}$$

b)

$$\begin{aligned}(x^3)^2 \cdot x^{-4} &= x^{3 \cdot 2} \cdot x^{-4} \\ &= x^{6-4} \\ &= x^2\end{aligned}$$

2. Evaluate, no calculator:

a) $27^{\frac{2}{3}} = \sqrt[3]{27^2} = \sqrt[3]{729} = 9$

b)

$$\begin{aligned}\left(\frac{1}{16}\right)^{\frac{-3}{4}} &= \left(\frac{16}{1}\right)^{\frac{3}{4}} \\ &= (2^4)^{\frac{3}{4}} = 2^3 \\ &= 8\end{aligned}$$

c) $4^{\frac{-1}{2}} = \sqrt[2]{4^{-1}} = \sqrt[2]{\frac{1}{4}} = \frac{1}{2}$

3. Expand and simplify:

a)

$$\begin{aligned}(x+3)(x-5) &= x^2 - 5x + 3x - 15 \\ &= x^2 - 2x - 15\end{aligned}$$

b)

$$\begin{aligned}(2x-1)^2 &= (2x-1)(2x-1) \\ &= 4x^2 - 2x - 2x + 1 \\ &= 4x^2 - 4x + 1\end{aligned}$$

4. Factorize:

- a) $x^2 - 9 = (x+3)(x-3)$
- b) $x^2 + 5x + 6 = (x+2)(x+3)$
- c) $6x^2 + x - 2 = (3x+2)(2x-1)$

5. Factorize Completely:

$$\begin{aligned}x^3 - x &= x(x^2 - 1) \\ &= x((x+1)(x-1))\end{aligned}$$

6. Solve, giving all solutions:

$$\begin{aligned}x^2 &= 5x \\ x &= \pm 5\end{aligned}$$

Attempt 2:

$$\begin{aligned}2x^2 &= 5x \\ 2x^2 - 5x &= 0 \\ x(2x - 5) &= 0\end{aligned}$$

7. Solve:

$$\begin{aligned}x^2 + 2x - 8 &= 0 \\ (x+4)(x-2) &= 0\end{aligned}$$

The quadratic has x intercepts at $(0, -2)$ and $(0, 4)$ Attempt 2: Assuming the intercepts are at $(-2, 0)$ and $(4, 0)$

$$(-2)^2 + 2(-2) - 8 = 0$$

$$4 - 4 - 8 \neq 0$$

My factorize term under expansion:

$$(x + 4)(x - 2) = 0$$

$$x^2 - 2x + 4x - 8 = 0$$

$$x^2 + 2x - 8 = 0$$

So the factorized term is correct, at second glance, coordinates should be $(-4, 0)$ and $(2, 0)$, testing:

$$(-4)^2 + 2(-4) - 8 = 0$$

$$16 - 8 - 8 = 0$$

$$0 = 0$$

So that one works, testing the next one:

$$(2)^2 + 2(2) - 8 = 0 \quad 4 + 4 - 8 = 0$$

$$0 = 0$$

Finding the y intercept:

$$y = (0)^2 + 2(0) - 8$$

$$y = 0 + 0 - 8$$

$$y = -8$$

So the y intercept is at $(0, -8)$

8. Make h the subject:

$$V = \frac{1}{3}\pi \cdot r^2 \cdot h$$

$$h = \frac{3V}{\pi \cdot r^2}$$

9. Solve simultaneously:

$$x + y = 7 \tag{1}$$

$$2x - 3y = 4 \quad (2)$$

$$(1)x + y = 7$$

$$x = -y + 7$$

$$y = -x + 7$$

Sub (1) into (2), first solving y , then x :

$$2(-y + 7) - 3y = 4$$

$$-2y + 14 - 3y = 4$$

$$-5y = -10$$

$$y = 2$$

$$2x - 3(-x + 7) = 4$$

$$2x + 3x - 21 = 4$$

$$5x = 25$$

$$x = 5$$

We can then confirm, first (1) then (2):

$$(5) + (2) = 7$$

$$7 = 7$$

and

$$2(5) - 3(2) = 4$$

$$10 - 6 = 4$$

$$4 = 4$$

10. Simplify: $\frac{x^2-4}{x-2} = x - 2$ where $x \neq 2$ Attempt 2: First factor $x^2 - 4$:

$$x^2 - 4 = (x + 2)(x - 2) \quad (3)$$

Then sub into original problem:

$$\frac{x^2 - 4}{x - 2} = \frac{(x + 2)(x - 2)}{x - 2}$$

$$\frac{(x + 2)\cancel{(x - 2)}}{\cancel{x - 2}} = x + 2$$

where $x \neq 2$

11. Express as a single fraction: $\frac{1}{x} + \frac{1}{x+1} = \frac{2}{2x+1}$

Attempt 2: $x \cdot (x+1) = x^2 + x$

$$\begin{aligned}\frac{1}{x} + \frac{1}{x+1} &= \frac{x+1}{x^2+x} + \frac{x}{x^2+x} \\ &= \frac{2x+1}{x^2+x}\end{aligned}$$

12. Rationalize the denominator: $\frac{2}{\sqrt[2]{3}-1} = \frac{4}{3+1} = 1$

Attempt 2, focusing on the denominator first:

$$(\sqrt[2]{3}-1)^2 = 3 - \sqrt[2]{3} - \sqrt[2]{3} + 1 \quad \text{not the one}$$

$$(\sqrt[2]{3}-1)(\sqrt[2]{3}+1) = 3 + \sqrt[2]{3} - \sqrt[2]{3} - 1 \quad \text{this is the one}$$

Now for the actual equation:

$$\frac{2}{\sqrt[2]{3}-1} = \frac{2}{\sqrt[2]{3}-1} \cdot \frac{\sqrt[2]{3}+1}{\sqrt[2]{3}+1}$$

$$\text{sub from above} = \frac{2(\sqrt[2]{3}+1)}{2}$$

$$\frac{\cancel{2}(\sqrt[2]{3}+1)}{\cancel{2}} = \sqrt[2]{3}+1$$

checking, subbing $\sqrt[2]{3}$ for 1.73:

$$\frac{2}{1.73-1} \approx 2.74$$

$$1.73+1 \approx 2.73 \approx 2.74$$